

Groupoid isotropy, the Baire Category Theorem, and Choice

Te Herenga Waka - OA Seminar

Clark, Thompson & Tolich

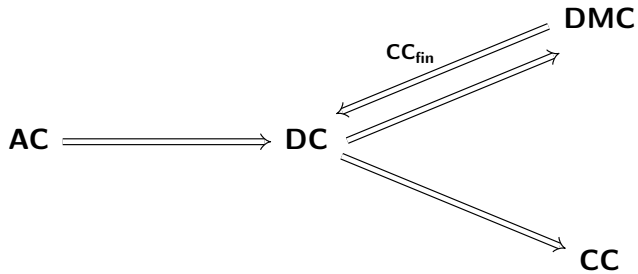
April 2, 2026

Groupoid
Isotropy

Baire Category
Theorem

Axioms of
Choice

1. **AC**: for every family $\{X_i\}_I$ of sets, the product $\prod_I X_i$ is non-empty.
2. **CC**: for every family $\{X_i\}_{\mathbb{N}}$ of sets, the product $\prod_{\mathbb{N}} X_i$ is non-empty.
3. **CC_{fin}**: for every family $\{X_i\}_{\mathbb{N}}$ of *finite* sets, the product $\prod_{\mathbb{N}} X_i$ is non-empty.
4. **DC**: for every nonempty set X and total relation $\mathcal{R}$ on X , there exists a sequence $(x_n)_{\mathbb{N}} \subseteq X$ such that $x_i \mathcal{R} x_{i+1}$ for all $i \in \mathbb{N}$.
5. **DMC**: for every nonempty set X and total relation $\mathcal{R}$ on X , there exists a family $\{X_i\}_{\mathbb{N}}$ of finite subsets of X such that for all $n \in \mathbb{N}$ and all $x \in X_n$, there exists $y \in X_{n+1}$ such that $x \mathcal{R} y$.



Definition

Let X be a topological space. We say that X is **Baire** if, whenever $\{D_n\}_{\mathbb{N}}$ is a collection of closed subsets of X each with empty interior, then $\bigcup_{\mathbb{N}} D_n$ has empty interior.

The following are *equivalences* shown by [HK00].

Every • is Baire	Axiom
totally bounded complete pseudometric space	CC
second countable complete pseudometric space	CC
complete pseudometric space	DC
complete metric space	DC
compact Hausdorff space	DMC

Groupoid
Isotropy

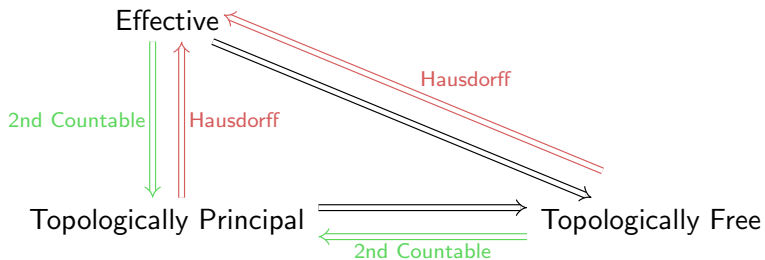
Baire Category
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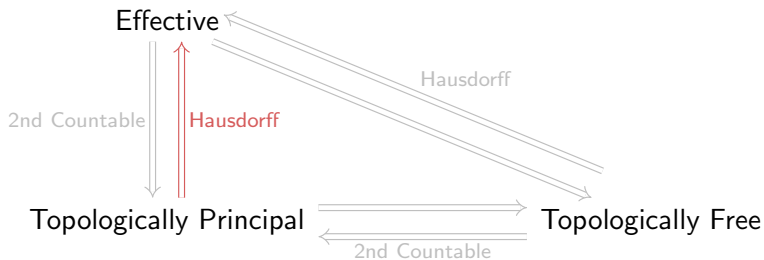
Axioms of
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Definition

Let $\mathcal{G}$ be an étale groupoid. We say that $\mathcal{G}$ is

1. *effective* if $\text{Int}(\text{Iso}(\mathcal{G})) = \mathcal{G}^{(0)}$;
2. *topologically principal* if $\{x \in \mathcal{G}^{(0)} : x\mathcal{G}x = \{x\}\}$ is dense in $\mathcal{G}^{(0)}$;
3. *topologically free* if $\text{Int}(\text{Iso}(\mathcal{G}) \setminus \mathcal{G}^{(0)}) = \emptyset$.





Consider $[0, 1]$ with the standard topology τ , and an adjoined point p . The *two-headed snake* is the groupoid

$$\mathcal{G} = [0, 1] \cup \{p\}$$

with $\mathcal{G}^{(2)} = \{(x, x) : x \in [0, 1]\} \cup \{(0, p), (p, 0), (p, p)\}$. We give $\mathcal{G}$ a topology by setting

$$\tau_{\mathcal{G}} := \tau \cup \{U \cup \{p\} - \{1\} : U \in \tau \text{ and } 1 \in U\}.$$

Then $\mathcal{G}$ is topologically principal since

$\{x \in \mathcal{G}^{(0)} : x\mathcal{G}x = \{x\}\} = (0, 1]$ which is dense in $[0, 1]$.

But, $\mathcal{G}$ is not effective since $\text{Int}(\text{Iso}(\mathcal{G})) = \mathcal{G} \neq \mathcal{G}^{(0)}$.

Theorem

[Ren08, Proposition 3.6(ii)] Suppose $\mathcal{G}$ is a topological groupoid with a countable cover of open bisections such that $\mathcal{G}^{(0)}$ is a Baire space. If $\mathcal{G}$ is effective, it is topologically principal.

Neighbourhood bisections: keep injectivity at the cost of openness.

Theorem

[BC14, Theorem 4.1] *The following are equivalent.*

1. *Every complete metric space is Baire ($\iff$ **DC**).*
2. *Let $\mathcal{G}$ be a topological groupoid with a countable cover of neighbourhood bisections such that $\mathcal{G}^{(0)}$ is a complete metric space. If $\mathcal{G}$ is effective, it is topologically principal.*

Definition

Let $\mathcal{G}$ be a topological groupoid. An open subset $U \subseteq \mathcal{G}$ is called an *open iso-section* if $r|_{U \setminus \mathcal{G}^{(0)}}$ is injective.

While an open bisection cannot both contain a unit x and an element of the fiber $x\mathcal{G}$, an open iso-section may contain both a unit and an element from its range fiber.

Clearly every open bisection is also an open iso-section.

Theorem

Let X be a topological space. The following are equivalent.

- 1. The space X is Baire.*
- 2. Let $\mathcal{G}$ be an étale groupoid with $\mathcal{G}^{(0)} = X$, and suppose $\mathcal{G}$ admits a countable cover of open iso-sections. If $\mathcal{G}$ is topologically free, then it is topologically principal.*

(1) $\implies$ (2)

Let $\mathcal{G}$ be a topologically free étale groupoid with $\mathcal{G}^{(0)}$ Baire. Let $\{B_n\}$ be a countable cover of $\mathcal{G}$ by open iso-sections.

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$$C_n := \overline{r(\text{Iso}B_n \setminus G^{(0)})}^0 \subseteq G^{(0)}.$$

One can show that each C_n has empty interior, and since $\mathcal{G}^{(0)}$ is Baire, $C := \bigcup C_n$ also has empty interior.

But C is exactly the units with nontrivial isotropy, and so $\mathcal{G}^{(0)} \setminus C$ consists of units with trivial isotropy, and is dense in $\mathcal{G}^{(0)}$.

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Let X be a space with basis τ . We want to show that X is Baire, so we construct a groupoid with $\mathcal{G}^{(0)} = X$.

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Let $\{C_n\}$ be a family of closed subsets of X with empty interior and let $C = \bigcup C_n$.

We define a groupoid $\mathcal{G} := X \cup \{p_x : x \in C\}$ such that $x\mathcal{G}x \cong \mathbb{Z}_2$ for $x \in C$, and $x\mathcal{G}x = \{x\}$ otherwise.

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As with the two-headed snake example, if $U \subseteq X$ is open and $x \in U \cap C$, then add a new basis element $U \setminus \{x\} \cup \{p_x\}$.

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For each n , set $B_n := \bigcup_{x \in C_n} X_x$. Then each B_n is an open iso-section, and the collection of such sets covers $\mathcal{G}$. Thus, by assumption, $\mathcal{G}$ is topologically principal.

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For each n , set $B_n := \bigcup_{x \in C_n} X_x$. Then each B_n is an open iso-section, and the collection of such sets covers $\mathcal{G}$. Thus, by assumption, $\mathcal{G}$ is topologically principal.

As before, the units with trivial isotropy is exactly $\mathcal{G}^{(0)} \setminus C$ which is dense, and so C has empty interior. Thus, X is Baire.

Groupoid
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Baire Category
Theorem

Axioms of
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Corollary

The following are equivalent:

1. every *second-countable complete pseudometric space* is Baire,
2. **CC**,
3. let G be an étale groupoid that has a countable cover of open iso-sections, such that $G^{(0)}$ is a *second-countable complete pseudometric space*. If G is topologically free, it is topologically principal.

Corollary

The following are equivalent:

1. every *complete metric space* is Baire,
2. **DC**,
3. let G be an étale groupoid that has a countable cover of open iso-sections, such that $G^{(0)}$ is a *complete metric space*. If G is topologically free, it is topologically principal.

Corollary

The following are equivalent:

1. every *compact Hausdorff space* is Baire,
2. **DMC**,
3. let G be an étale groupoid that has a countable cover of open iso-sections, such that $G^{(0)}$ is a *compact Hausdorff space*. If G is topologically free, it is topologically principal.

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For more - read our preprint! [[CTT26](#)]





Jonathan Brown and Lisa Orloff Clark.

A groupoid formulation of the Baire Category Theorem.

[Fundamenta Mathematicae](#), 226(2):123–130, 2014.



Lisa Orloff Clark, Ryan Thompson, and Ilija Tolich.

Topologically free non-Hausdorff groupoids, February 2026.



Horst Herrlich and Kyriakos Keremedis.

The Baire Category Theorem and choice.

[Topology and its Applications](#), 108(2):157–167, December 2000.



Jean Renault.

Cartan subalgebras in C^* -algebras.

[Irish Mathematical Society Bulletin](#), 2008.